Seat No.: ________ Enrolment No.___________

# GUJARAT TECHNOLOGICAL UNIVERSITY

**BE - SEMESTER– VI (NEW) EXAMINATION – WINTER 2021**

**Subject Code:3160917 Date:02/12/2021**

**Subject Name:Wind And Solar Energy**

**Time:10:30 AM TO 01:00 PM Total Marks: 70**

**Instructions:**

1. **Attempt all questions.**
2. **Make suitable assumptions wherever necessary.**
3. **Figures to the right indicate full marks.**
4. **Simple and non-programmable scientific calculators are allowed.**

| | | | **MARKS** |
|---|---|---|---|
| **Q.1** | **(a)** | Define Cut in speed, Cut out speed and Tip speed ratio | **03** |
| | **(b)** | Write Betz Law and mention betz limit value in case of wind turbine Wind | **04** |
| | **(c)** | Classify and Explain Fixed and Variable speed wind turbines | **07** |
| **Q.2** | **(a)** | List the types of generator used in wind power plant | **03** |
| | **(b)** | State role of Power electronics converter in wind power | **04** |
| | **(c)** | Explain construction and working of Doubly-Fed Induction Generators with characteristics | **07** |
| | | **OR** | |
| | **(c)** | Explain Converter control techniques in case of Wind power | **07** |
| **Q.3** | **(a)** | Define zenith, solar altitude and azimuth angle in solar geometry | **03** |
| | **(b)** | Prove that solar day length is function of latitude and declination angle | **04** |
| | **(c)** | Explain structure of solar cell, module, panel and array | **07** |
| | | **OR** | |
| **Q.3** | **(a)** | Draw and explain V-I characteristics of a PV cell | **03** |
| | **(b)** | Compare monocystaline, polycrystalline and thin film type solar module. | **04** |
| | **(c)** | What is Maximum Power Point Tracking (MPPT) system and which types of algorithms used to track maximum power from solar PV System. | **07** |
| **Q.4** | **(a)** | Differentiate Grid-Connected System and Standalone system | **03** |
| | **(b)** | How solar water pump works? | **04** |
| | **(c)** | Explain real and reactive power regulation in case of wind power | **07** |
| | | **OR** | |
| **Q.4** | **(a)** | Mention voltage and frequency operating limits on integration of solar and wind. | **03** |
| | **(b)** | List out Power quality issues during integration of solar –wind with grid. | **04** |
| | **(c)** | Write technical note on Solar Refrigeration and Air Conditioning | **07** |
| **Q.5** | **(a)** | How Solar water heater works? | **03** |
| | **(b)** | Give the type of Solar Concentrators used in solar thermal plant | **04** |
| | **(c)** | Explain working of solar thermal plant | **07** |
| | | **OR** | |
| **Q.5** | **(a)** | What is solar pond? | **03** |
| | **(b)** | What is grid code and why it is required? | **04** |
| | **(c)** | List out application of solar –thermal systems | **07** |

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